example, trillions of bodies each a kilometer in size would be needed to coalesce to make a planet with the mass of the Earth. Once there were much larger numbers of worldlets in the planetary part of the Solar System. Most of them are now gone—ejected into interstellar space, fallen into the Sun, or sacrificed in the great enterprise of building moons and planets. But out beyond Neptune and Pluto the discards, the leftovers that were never aggregated into worlds, may be waiting—a few largish ones in the 100-kilometer range, and stupefying numbers of kilometer-sized and smaller bodies peppering the outer Solar System all the way out to the Oort Cloud.

In this sense there *are* planets beyond Neptune and Pluto—but they are not nearly as big as the Jovian planets, or even Pluto. Larger worlds may, for all we know, also be hiding in the dark beyond Pluto, worlds that can properly be called planets. The farther away they are, the less likely it is that we would have detected them. They cannot lie just beyond Neptune, though; their gravitational tugs would have perceptibly altered the orbits of Neptune and Pluto, and the *Pioneer 10* and *11* and *Voyager 1* and *2* spacecraft.

The newly discovered cometary bodies (with names like 1992QB and 1993FW) are not planets in this sense. If our detection threshold has just encompassed them, many more of them probably remain to be discovered in the outer Solar System—so far away that they're hard to see from Earth, so distant that it's a long journey to get to them. But small, quick ships to Pluto and beyond are within our ability. It would make good sense to dispatch one by Pluto and its moon Charon, and then, if we can, to make a close pass by one of the denizens of the Kuiper Comet Belt.

The rocky Earthlike cores of Uranus and Neptune seem to have accreted first, and then gravitationally attracted massive amounts of hydrogen and helium gas from the ancient nebula out of which the planets formed. Originally, they lived in a hailstorm. Their gravities were just sufficient to eject icy worldlets, when they came too close, far out beyond the realm of the planets, to populate the Oort Comet Cloud. Jupiter and Saturn became gas giants by the same process.